01

National University of Computer and Emerging Sciences

FAST School of Computing | Fall-2023 | Islamabad Campus

MT-1003: Calculus & Analytical Geometry

CS

Tuesday, 26th September, 2023

Serial No:

Sessional Exam-I

Total Time: 1 Hour

Total Marks: 45

Signature of Invigilator

Course Instructors

Dr. Hamda Khan, Mr. Usman Rashid, Dr. Imran Shahzad

[redacted] Student Name | [redacted] Roll No. | [redacted] Course Section | [redacted] Student Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

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Question 1 [12 Marks]

Four feet of wire is to be used to form a square (having side 'x') and a circle (of radius 'r'). Determine a function that expresses the combined area of both figures, then express the area in terms of 'x'. What is the domain of this function?

sides of square $= x$

Perimeter of square $= 4x$ $(f(x))$

radius of circle $= r$

Circumference of circle $= 2\pi r$ $(g(x))$

As same wire is used so perimeter of square is equal to circumference of circle.

$$4x = 2\pi r \Rightarrow r = \frac{4x}{2\pi} = \frac{2x}{\pi}$$

$$\boxed{r = \frac{2x}{\pi}}$$

4.5

Now

Area of square $= x^2$

Area of circle $= \pi r^2 = \pi \left(\frac{2x}{\pi}\right)^2 = 4x^2\pi$

Combined Area = A. of square + A. of circle

$$(f+g)(x) = x^2 + 4x^2\pi$$

Domain $= \mathbb{R}$ or $(-\infty, +\infty)$

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Question 2 [13 Marks]

A toy company wants to design a track for a toy car that starts out along a parabolic curve and then converts to a straight line. The function that describes the track is to have the form:

$$f(x) = \begin{cases} \frac{1}{5}x^2 + bx + c & if\ x < 4 \\ \frac{1}{3}x + 2 & if\ x \geq 4 \end{cases}$$

where x and $f(x)$ are in inches. For the car to move smoothly along the track, the function $f(x)$ must be both continuous and differentiable at 4. Find values of b and c that make $f(x)$ both continuous and differentiable. (Note: Find derivative by definition).

For a function to be continuous: L.H.L = R.H.L = f(4)

$$\frac{1}{5}x^2 + bx + c = \frac{1}{3}x + 2$$

$$c = -\frac{1}{5}x^2 + \frac{1}{3}x - bx + 2 \quad = -\frac{1}{5}x^2 + \left(\frac{1}{3} - b\right)x + 2$$

$$c = -\frac{1}{5}x^2 + \left(\frac{1-3b}{3}\right)x + 2$$

Apply $\lim_{x \to 4}$

$$c = -\frac{1}{5}(4)^2 + \left(\frac{1-3b}{3}\right)4 + 2$$

$$c = -\frac{16}{5} + \frac{4-12b}{3} + 2$$

1.5

Now taking differential of $f(x) = \frac{1}{5}x^2 + bx + c$ by definition.

$$\lim_{h \to 4} f(x) = \frac{f(x+h) - f(x)}{h}$$

=

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Question 3A [05 Marks]

Graph the rational function. Include the graphs and the equations of the asymptotes.

$$y = \frac{x^2-1}{2x+4}.$$

* No horizontal asymptote possible because the degree of denominator is smaller than the numerator.
* It has a vertical asymptote at $x = -2$.

Graph of $y = \frac{x^2-1}{2x+4}$

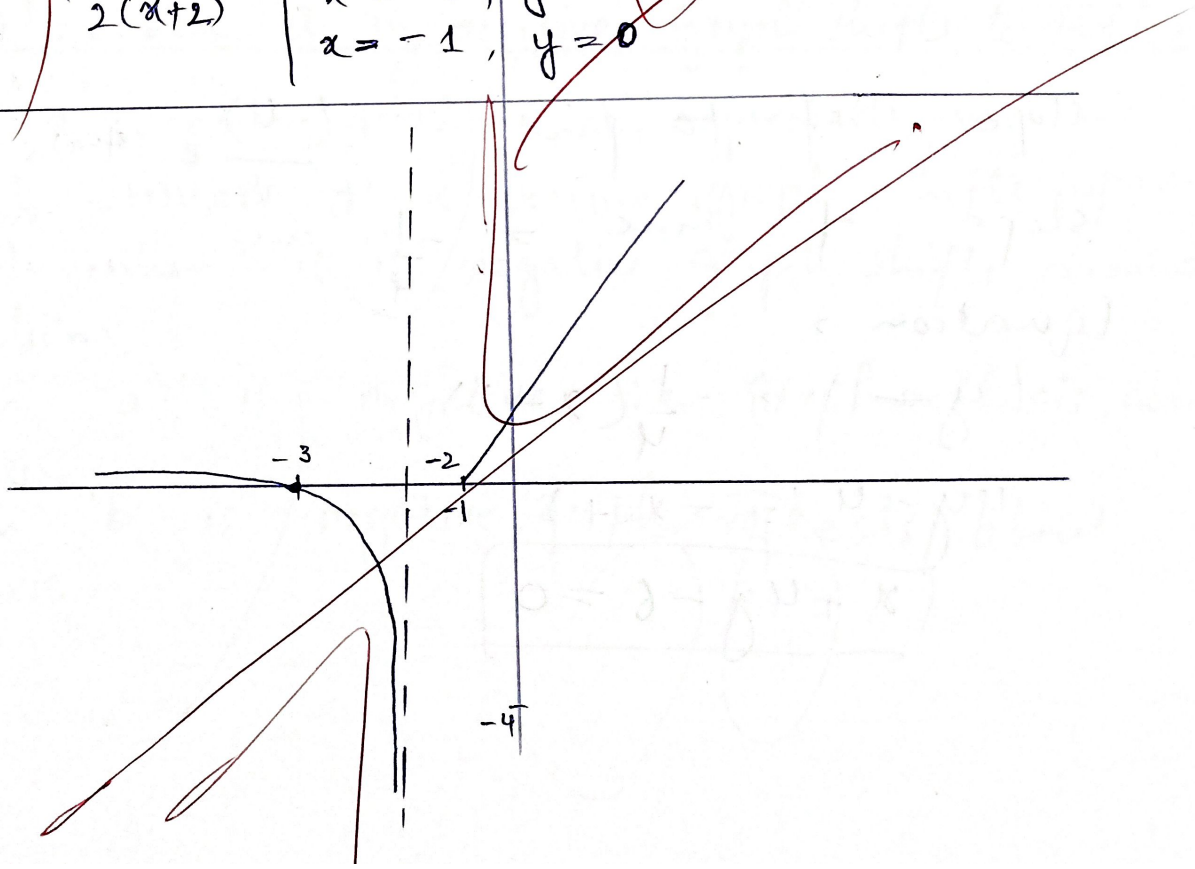
$= \frac{(x+1)(x-1)}{2(x+2)} = \frac{x^2-1-3+3}{2x+4}$

$= \frac{x^2-4+3}{2x+4} = \frac{(x-2)(x+2)+3}{2(x+2)} = \frac{(x-2)(x+2)}{2(x+2)} + \frac{3}{2(x+2)}$

$= \frac{x-2}{2} + \frac{3}{2(x+2)}$

$x = -3, \; y = -4$

$x = -1, \; y = 0$



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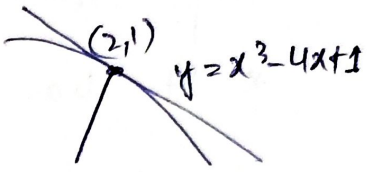
Question 3B [04 Marks]

Find an equation for the line perpendicular to the tangent to the curve

$$y = x^3 - 4x + 1.$$

at the point (2,1).

for perpendicular line $m = -\frac{1}{m}$



slope of tangent $= -\frac{(4)}{1}$

$= 0$

slope of line $= \infty$

Equation of line

$(y - 1) = \infty (x - 2)$

or

slope of tangent $= -\frac{(-4)}{1} = 4$

slope of line $= -\frac{1}{4}$

Equation :

$(y - 1) = -\frac{1}{4}(x - 2)$

$4y - 4 = -x + 2$

$$x + 4y - 6 = 0$$

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Question 3C [04 Marks]

Elaborate the rules of shifting, stretching, compressing, and reflecting the graph of function in terms of symbols a, b, c, and d. Explain the role of each one.

$$y = af(b(x + c)) + d.$$

1. In the given equation 'a' represents vertical compresion or stretch. When 'a' increases, vertical stretch increases and vice verse.
2. In the given equation 'b' represents horizontal stretch or compress. When 'b' increases horizontal compress increases.
3. In the given equation 'c' represents horizontal shift. When 'c' is negative, graph shifts to right and when it is positive graph shifts to left.
4. In the given equation 'd' represents vertical shift. When 'd' is positive graph shifts upwards and when it is negative graph shifts downwards.

Reflection:

1. When 'a' is negative the graph reflects about x-axis.
2. When 'b' is negative, graph reflects about y-axis.

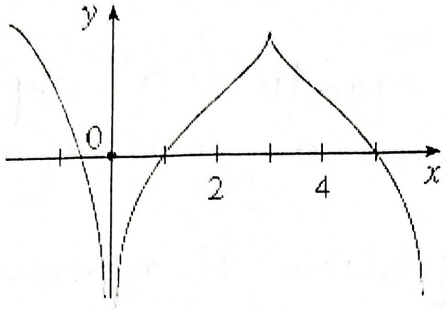
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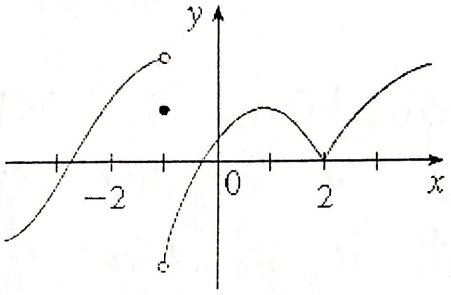
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Question 4 [07 Marks]

The graphs of two functions are given. State, with the reason, the point(s) where given functions are not differentiable.





6.5

In figure (1) graph is not ~~defined~~ differentiable at $x = 0$ due to vertical asymptotes, and at $x = 3$ due to formation of cusp at that point.

In figure (2) graph is not ~~defined~~ differentiable at -1 because it is ~~unde~~ discontinuous at -1, and at $x = 2$ due to a cusp.

Corner

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MT-1003: Calculus and Analytical Geometry (CS)

Serial No:

2nd Sessional Exam

Total Time: 1 Hour

Total Marks: 45

Tuesday, 7th November, 2023

Course Instructors

Dr. Hamda Khan, Mr. Usman Rashid, Dr. Imran Shahzad

Signature of Invigilator

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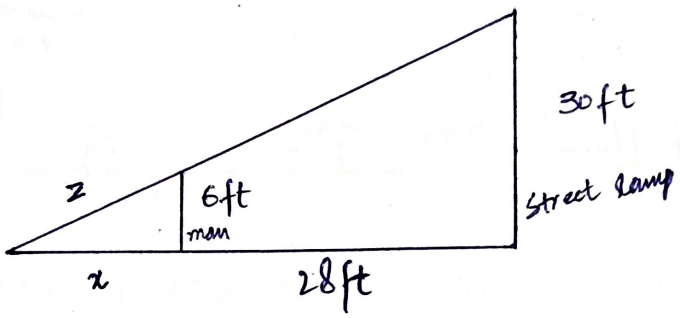
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Question 1 [12 Marks]

A man, who is 28ft away from a 30ft tall streetlamp is sinking into quicksand. At the moment when 6ft of him are above the ground, his height above the ground is shrinking at a rate of 2 ft/s.

a) How long will the man's shadow be at the moment when 6 feet of him are above the ground?

b) At what rate is the length of the man's shadow changing at the moment 6ft of him are above the ground? Is his shadow growing or shrinking at that moment?



$y = 6ft$, $\frac{dy}{dt} = 2ft/s$

a) Using the phenomenon of similar triangles

$$\frac{6}{x} = \frac{30}{28+x} \Rightarrow 168 + 6x = 30x$$

$$168 = 30x - 6x \Rightarrow 24x = 168 \Rightarrow x = \frac{168}{24} = 7ft$$

So when the man is 6ft above the ground his shadow will be of 7ft.

b) In the figure

$$z^2 = y^2 + x2 \Rightarrow z = \sqrt{y^2 + x^2} = \sqrt{6^2 + 72} = 9.22ft$$

Now using Pythagorus theorem, we will find $\frac{dx}{dt}$.

$$z^2 = y^2 + x^2$$

$(9.22)^2 = y^2 + x^2$

Differentiate w.r.t x.

$$0 = 2y\frac{dy}{dt} + 2x\frac{dx}{dt}$$

$$2x\frac{dx}{dt} = -2y\frac{dy}{dt} \Rightarrow \frac{dx}{dt} = \frac{-2y}{2x}\cdot\frac{dy}{dt}$$

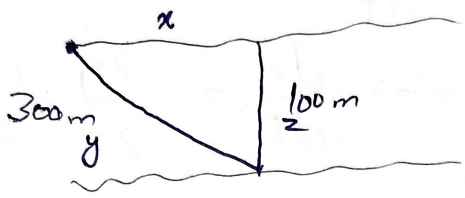
$$\frac{dx}{dt} = \frac{-y}{x}\cdot\frac{dy}{dt} = \frac{-6}{7}\cdot 2 = -1.714 \text{ ft/s}$$

09

Here we can see that the mans shadow is shrinking.

Question 2 [13 Marks]

You are standing on the bank of a river that is 100m wide and see someone needing help 300m up the opposite shore. You can swim at 3m/s and run at 5m/s, and you want to get to the person as quickly as possible. To what point on the opposite shore should you swim, before running the rest of the way?



$V_{swim} = 3\ m/s$, $V_{run} = 5\ m/s$

$y^2 = x^2 + z^2$ $\quad (300)^2 = x^2 + z^2$

differentiate w.r.t t

$0 = 2x\frac{dx}{dt} + 2z\frac{dz}{dt}$

$2x\frac{dx}{dt} = -2z\frac{dz}{dt}$

$x = \frac{-2z\frac{dz}{dt}}{2\frac{dx}{dt}} = \frac{-2(100)(3)}{2(5)}$

0.3

$x = -60m$

You should swim to a point 60 m away from the person and then run for help.

Question 3A [07 Marks]

Find $\frac{dy}{dx}$, if $y = 2^{\cos x + \ln x}$.

$$\frac{dy}{dx} = 2^{\cos x + \ln x} \cdot \ln(2) \cdot \frac{d}{dx}(\cos x + \ln x)$$

$$= 2^{\cos x + \ln x} \cdot \ln(2) \cdot \left(-\sin x + \frac{1}{x}\right)$$

$$= 2^{\cos x + \ln x} \cdot \ln(2) \left(\frac{1}{x} - \sin x\right)$$

how?

Question 3B [06 Marks]

Find $\lim_{x \to 0} (1 + \sin x)^{1/x}$.

Let $y = \lim_{x \to 0} (1 + \sin x)^{1/x}$

Taking ln on both sides

$\ln y = \ln (1+\sin x)^{1/x}$

$\ln y = \frac{1}{x} \cdot \ln (1+\sin x)$ 01

Differentiate w.r.t x

$\frac{1}{y} \cdot \frac{dy}{dx} = \left[\frac{1}{x} \cdot \frac{1}{1+\sin x}\right] + \ln(1+\sin x) \cdot \frac{-1}{x^2}$

$\frac{dy}{dx} = y\left[\frac{1}{x} \cdot \frac{1}{1+\sin x}\right] - \frac{\ln(1+\sin x)}{x^2}$

=

Question 4 [07 Marks]

Sketch the graph of $f(x) = \frac{x^2+1}{x}$ by finding:

(a) All the critical points.
(b) Intervals for increasing and decreasing.
(c) Intervals for concave up and concave down.

Domain = $(-\infty, 0) \cup (0, \infty)$

Vertical asymptote at $x = 0$

slanting asymptote: $x \overline{)\frac{x^2+1}{x}}$, $\frac{x^2}{1}$

for critical points $f'(x) = 0$, $f'(x) = \frac{x(2x) - (x^2+1)}{x^2} = \frac{2x^2 - x^2 - 1}{x^2} = \frac{x^2-1}{x^2}$

$\frac{x^2-1}{x^2} = 0 \Rightarrow x^2 - 1 = 0$, $(x-1)(x+1) = 0 \Rightarrow x = 1, x = -1$

$x \neq 0$

Intervals	$-\infty < x < -1$	$-1 < x < 0$	$0 < x < 1$	$1 < x < \infty$
value of $f(x)$	+	−	−	+
	increasing	decreasing	~~increasing~~ decreasing	~~decreasing~~ increasing

for points of inflection: $f''(x) = \frac{x^2(2x) - (x^2-1)(2x)}{x^4} = \frac{2x^3 - 4x^2 + 2x}{x^4}$

$f''(1) = 0$, $f''(-1) = -8$ (concave down)

C.U ??

C.D ??

